

LECTURE 3 · WEB TECHNOLOGIES

HTML

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HTML Style Attribute

- The HTML style attribute is used to add styles to an element, such as color, font, size, and more.
- With HTML styles, styles can be added to HTML elements directly by using the style attribute, or indirectly in separate style sheets (CSS files).

<*tagname style*="property:value;">

<*h1 style*="color:red; font-size:30px;">This is a heading</*h1*>

Tables in HTML

- The HTML table model allows authors to arrange data -- text, preformatted text, images, links, forms, form fields, other tables, etc. -- into rows and columns of cells.
- Each table may have an associated caption that provides a short description of the table's purpose.
- A table is divided into rows, and each row is divided into cells. Table cells may either contain "header" information (TH element) or "data" (TD element).

Common HTML Table Elements and Attributes

HTML Table Elements	Description
<table>...</table>	The Table delimiter.
<tr ...>...</tr>	Used to specify number of rows in a table.
<td ...>...</td>	Specifies table data cells
<th ...>...</th>	Table Header cell
<caption ...>...</caption>	Specifies the table Caption.
<thead>...</thead>	Specifies the Table head.
<tbody>...</tbody>	Specifies the Table body.
<tfoot>...</tfoot>	Specifies the Table footer.

</> HTML Code

Copy Code

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <title>Creating Tables in HTML</title>
5  </head>
6  <body>
7      <h2>HTML Table (Default Style)</h2>
8      <table>
9          <tr>
10             <th>No.</th>
11             <th>Name</th>
12             <th>Age</th>
13         </tr>
14         <tr>
15             <td>1</td>
16             <td>Peter Parker</td>
17             <td>16</td>
18         </tr>
19         <tr>
20             <td>2</td>
21             <td>Clark Kent</td>
22             <td>34</td>
23         </tr>
24         <tr>
25             <td>3</td>
26             <td>Harry Potter</td>
27             <td>11</td>
28         </tr>
29     </table>
30 </body>
31 </html>
```

Heading

Table Start

Table Header

Row 1

Row 2

Row 3

Table End

Tables in HTML

- Tables do not have any borders by default. You can use the *CSS border* property to add borders to the tables. Also, table cells are sized just large enough to fit the contents by default.
- To add more space around the content in the table cells you can use the *CSS padding* property. Each table may have an associated caption that provides a short description of the table's purpose.

index.html — Copy Code

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Adding Borders and Paddings
      to HTML Tables</title>
    <style>
        table, th, td {
            border: 1px solid black;
        }
        th, td {
            padding: 10px;
        }
    </style>
</head>
<body>
    <h2>Table with Separated Borders</h2>
    <table>
        <tr>
            <th>No. </th>
            <th>Name</th>
            <th>Age</th>
        </tr>
        <tr>
            <td>1</td>
            <td>Peter Parker</td>
            <td>16</td>
        </tr>
        <tr>
            <td>2</td>
            <td>Clark Kent</td>
            <td>34</td>
        </tr>
        <tr>
            <td>3</td>
            <td>Harry Potter</td>
            <td>11</td>
        </tr>
    </table>
</body> </html>
```

Adds 1px solid border to table cells and headers

Adds 10px padding inside each cell

Table Header

Row 1

Row 2

Row 3

Live Preview

Table with Separated Borders

No.	Name	Age
1	Peter Parker	16
2	Clark Kent	34
3	Harry Potter	11

What's happening?

- Each cell `(th, td)` has a 1px solid black border.
- Each cell has 10px padding, creating space between content and border.

Understanding Borders and Padding

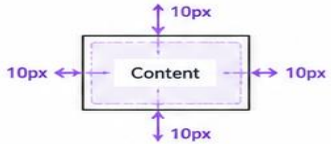
BORDER (1px solid black)

The border creates the visible outline around each cell.



PADDING (10px)

Padding adds space inside the cell, between content and border.



Tip

Adjust the padding value to increase or decrease the space inside cells.

<table>...</table>

<table>...</table> Attributes	Description
background	Specifies a background image for the table. If used in the <table> element, the image in question will be tiled behind all of the table cells.
bgcolor="#rrggbb|color name"	Specifies the background color of the table to be specified, using either the specified color names, or a rrggbb hex triplet.
align="left|right"	It allows a table to be aligned to the left or right of the page, allowing text to flow around the table.
border	Control and set the borders to be displayed for the table
bordercolor="#rrggbb|color name"	sets the border color of the table, using either the specified color names, or a rrggbb hex triplet. It is necessary for the border attribute to be present in the main <table> element for border coloring to work.
cellpadding=value	Specifies the amount of white space between the borders of the table cell and the actual cell data.

<tr ...>...</tr>

- This stands for table row.

<tr>...</tr> Attributes	Description
bgcolor="#rrggbb|color name"	Specifies the background color of the table row to be specified, using either the specified color names, or a rrggbb hex triplet.
align="left|center|right"	Specifies whether text inside the table cell(s) is aligned to the left, right or center of the cell.
valign="top|middle|bottom|baseline"	controls whether text inside the table cell(s) is aligned to the top, bottom, or vertically centered within the cell.
bordercolor="#rrggbb|color name"	sets the border color of the table row, using either the specified color names, or a rrggbb hex triplet. It is necessary for the BORDER attribute to be present in the main <TABLE> element for border coloring to work.

<td ...>...</td>

<td>...</td> Attributes	Description
background	Internet Explorer supports the placing of images inside the <td> element. If used in the <td> element, the image in question will be tiled behind the particular data cell.
bgcolor="#rrggbb|color name"	Specifies the background color of the data cell to be specified, using either the specified color names, or a rrggbb hex triplet.
align="left|center|right"	Specifies text alignment inside the table cell(s).
valign="top|middle|bottom|baseline"	controls whether text inside the table cell(s) is aligned to the top, bottom, or vertically centered within the cell.
bordercolor="#rrggbb|color name"	sets the border color of the data cell. Necessary for the border attribute to be present in the main <table> element for border coloring to work.
rowspan="value"	This attribute can appear in any table cell (<th> or <td>) and specifies how many rows of the table this cell should span. The default ROWSPAN for any cell is 1.
colspan="value"	This attribute can appear in any table cell (<th> or <td>) and it specifies how many columns of the table this cell should span. The default colspan for any cell is 1.

<th ...>...</th>

- This stands for table header.
- Header cells are identical to data cells in all respects, with the exception that header cells are in a bold font, and have a center default align.
- <th ...>...</th> has the same attribute as the <td ...>...</td> element.

<caption>...</caption>

- This represents the caption for a table.
- <CAPTION> elements should appear inside the <TABLE> but not inside table rows or cells.
- The caption accepts an alignment attribute that defaults to ALIGN=top but can be explicitly set to ALIGN=bottom.
- Like table cells, any document body HTML can appear in a caption.
- Captions are, by default horizontally centered with respect to the table, and they may have their lines broken to fit within the width of the table.

<caption>...</caption>

- The <caption> element can accept the following attributes:

align="top|bottom”: Specifies whether the caption appears above or below the table, using the top and bottom values, defaulting to top.

- Example

```
<table border="1">
  <caption align="bottom">My savings</caption>
  <tr>
    <th>Month</th>
    <th>Savings</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
  </tr>
</table>
```

Lists

- HTML supports several mechanisms for specifying lists of information.
- In HTML there are 3 different types of lists, all of which may be nested.
- If used they should be present in the <body> of an HTML document.
- Common HTML list elements are:
  - : unordered list; bullets
  - : ordered list; numbers
  - <dl> : definition list; dictionary

HTML Unordered Lists

- An unordered list created using the <ul> element, and each list item starts with the <li> element.

HTML Code (Copy Code)

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <title>HTML Unordered List</title>
5  </head>
6  <body>
7      <!-- Simple Unordered List -->
8      <h2>HTML Unordered List</h2>
9      <ul>
10         <li>Chocolate Cake</li>
11         <li>Black Forest Cake</li>
12         <li>Pineapple Cake</li>
13     </ul>
14     <hr>
15     <!-- Nested Unordered List -->
16     <h2>HTML Nested Unordered List</h2>
17     <ul>
18         <li>Chocolate Cake
19             <ul>
20                 <li>Chocolate Velvet Cake</l
21                 <li>Chocolate Lava Cake</li
22             </ul>
23         </li>
24         <li>Black Forest Cake</li>
25         <li>Pineapple Cake</li>
26     </ul>
```

Document Type

Unordered List (3 items)

Nested Unordered List

Live Preview

HTML Unordered List

- Chocolate Cake
- Black Forest Cake
- Pineapple Cake

HTML Nested Unordered List

- Chocolate Cake
  - Chocolate Velvet Cake
  - Chocolate Lava Cake
- Black Forest Cake
- Pineapple Cake

- `<ul>` : Unordered list (uses bullet points)
- `<li>` : List item (inside `<ul>`)
- `<ul>` inside `<li>` : Nested list

HTML Unordered Lists

- You can also change the bullet type in your unordered list using the CSS *list-style-type* property. The following style rule changes the type of bullet from the default disc to square:

HTML & CSS Code — Copy Code

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <title>Change Bullet Type in an
      HTML Unordered List Using CSS</title>
5    <style>
6      ul {
7        list-style-type: square;
8      }
9    </style>
10 </head>
11 <body>
12   <ul>
13     <li>Chocolate Cake</li>
14     <li>Black Forest Cake</li>
15     <li>Pineapple Cake</li>
16   </ul>
17 </body>
18 </html>
```

CSS Rule
Sets the bullet style for all `<ul>` items

Unordered List
List items will appear with square bullets

Live Preview — Light | Fullscreen

Output with `list-style-type: square;`

- Chocolate Cake
- Black Forest Cake
- Pineapple Cake

Try other bullet types:

disc (default)	circle ○	square ■ _	none —	decimal 1.

More options: `lower-alpha`, `upper-roman`, etc.

HTML Ordered Lists

- An ordered list created using the <ol> element, and each list item starts with the <li> element.
- The OL element has ***type*** attribute which allows authors to specify whether the list items should be marked with.
- It has following values:
  - <ol type=a> Small letters. For example a, b, c ...
  - < ol type =A> Capital letters. For example A, B, C ...
  - < ol type =i> Small roman numerals. For example i, ii, iii ...
  - < ol type =I> Large roman numerals. For example I, II, III ...
  - < ol type =1> The default numbers. For example 1, 2, 3 ...

HTML Code

Copy Code

```
<!DOCTYPE html>
<html lang="en">
<head>
   <title>HTML Ordered List</title>
</head>
<body>
   <h2>>HTML Ordered List</h2>
   <ol>
      <li>Fasten your seatbelt</li>
      <li>Starts the car's engine</li>
      <li>Look around and go</li>
   </ol>
   <hr>
   <h2>>HTML Nested Ordered List</h2>
   <ol>
      <li>Fasten your seatbelt</li>
      <li>Starts the car's engine</li>
      <li>Look around and go
        <ol>
          <li>Check the blind spot</li>
          <li>Check surrounding for safety</li>
        </ol>
      </li>
   </ol>
</body>
</html>
```

Document Structure

Ordered List

Horizontal Rule

Nested Ordered List

Live Preview

Light Fullscreen

HTML Ordered List

1. Fasten your seatbelt
2. Starts the car's engine
3. Look around and go

HTML Nested Ordered List

1. Fasten your seatbelt
2. Starts the car's engine
3. Look around and go
    1. Check the blind spot
    2. Check surrounding for safety

How it Works

- ✓ `<ol>` creates an ordered list (numbers by default)
- ✓ `<li>` defines each list item
- ✓ You can nest an `<ol>` inside an `<li>` to create sub-lists

HTML Ordered Lists

- The ol element has start attribute which allows authors to start at values other than 1.
- Thus *start*=5 would display either an 'E', 'e', 'V', 'v', or '5' based on the *type* attribute.
- Example

```
<h4 align="center">Goals</h4>
<ol start="4" >
<li>Buy Food</li>
<li>Enroll in College</li>
<li>Get a Degree</li>
</ol>
```

HTML Description Lists

- A description list is a list of items with a description or definition of each item.
- The description list is created using <dl> element. The <dl> element is used in conjunction with the <dt> element which specify a term, and the <dd> element which specify the term's definition.
- Browsers usually render the definition lists by placing the terms and definitions in separate lines, where the term's definitions are slightly indented.

HTML Code

Copy Code

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4     <title>HTML Description or Definition List</title>
5  </head>
6  <body>
7     <h2>HTML Definition List</h2>
8     <dl>
9        <dt>Bread</dt>
10       <dd>Baked food made of flour.</dd>
11       <dt>Coffee</dt>
12       <dd>A drink made from roasted coffee beans.</dd>
13    </dl>
14 </body>
15 </html>
```

Heading

Definition List

Term

Description

Term

Description

Live Preview

Light

Fullscreen

HTML Definition List

Bread

Baked food made of flour.

Coffee

A drink made from roasted coffee beans.

How it Works

`<dl>` - Defines the definition list

`<dt>` - Defines the term (name)

`<dd>` - Defines the description of the term

HTML Iframes

- An iframe is used to display a web page within a web page.
- *<iframe src*="url"*></iframe>*
- The URL points to the location of the separate page.
- The height and width attributes are used to specify the height and width of the iframe.

  <iframe src="http://www.google.com" *width*="200" *height*="200"*></iframe>*
- The frameborder attribute specifies whether or not to display a border around the iframe. Set the attribute value to "0" to remove the border

HTML Forms

- HTML Forms are used to select different kinds of user input and to pass data to a server.
- A form can contain input elements like text fields, checkboxes, radio-buttons, submit buttons and more.
- A form can also contain select lists, textarea, fieldset, legend, and label elements.
- A form will take input from the viewer and depending on your needs; you may store that data into a file, place an order, gather user statistics, register the person to your web forum, or maybe subscribe them to your weekly newsletter.

HTML Forms

- The following elements are used to create forms:

  - <form>...</form> : A form within a document
  - <input ...>...</input> : One input field
  - <option> : One option within a Select element
  - <select>...<select> : A selection from a finite set of options
  - <textarea ...>...</textarea> : A multi line input field

Frequently Used Form Attributes

- The following table lists the most frequently used form element's attributes:

Attribute	Description
name	Specifies the name of the form.
action	Specifies the URL of the program or script on the web server that will be used for processing the information submitted via form.
method	Specifies the HTTP method used for sending the data to the web server by the browser. The value can be either get (the default) and post.
target	Specifies where to display the response that is received after submitting the form. Possible values are _blank, _self, _parent and _top.
enctype	Specifies how the form data should be encoded when submitting the form to the server. Applicable only when the value of the method attribute is post.

<form>...</form>

- The <form> tag is used to create an HTML form. Here's a simple example of a login form:

```
<form>
    <label>Username: <input type="text"></label>
    <label>Password: <input type="password"></label>
    <input type="submit" value="Submit">
</form>
```

The <input> Element

- The HTML <input> element is the most used form element.
- An <input> element can be displayed in many ways, depending on the type attribute.

Type	Description
<input type="text">	Displays a single-line text input field
<input type="radio">	Displays a radio button (for selecting one of many choices)
<input type="checkbox">	Displays a checkbox (for selecting zero or more of many choices)
<input type="submit">	Displays a submit button (for submitting the form)
<input type="button">	Displays a clickable button

Text Fields

- The <input type="text"> defines a single-line input field for text input.
- Single-line text input controls are created using an <input> element, whose type attribute has a value of text. Here's an example of a single-line text input used to take username:

```
<form>
  <label for="username">Username:</label>
  <input type="text" name="username" id="username">
</form>
```

Password Field

- Password fields are similar to text fields. The only difference is; characters in a password field are masked, i.e. they are shown as asterisks or dots.
- This is to prevent someone else from reading the password on the screen. This is also a single-line text input controls created using an <input> element whose type attribute has a value of password.

```
<form>
<label for="user-pwd">Password:</label>
<input type="password" name="user-password“ id="user-pwd">
</form>
```

Radio Buttons

- Radio buttons are used to let the user select exactly one option from a pre-defined set of options.
- It is created using an <input> element whose type attribute has a value of radio.

```
<form>
    <input type="radio" name="gender" id="male">
    <label for="male">Male</label>
    <input type="radio" name="gender" id="female">
    <label for="female">Female</label>
</form>
```

Checkboxes

- Checkboxes allows the user to select one or more option from a pre-defined set of options.
- It is created using an <input> element whose type attribute has a value of checkbox.

```
<form>
    <input type="checkbox" name="sports" id="soccer">
    <label for="soccer">Soccer</label>
    <input type="checkbox" name="sports" id="cricket">
    <label for="cricket">Cricket</label>
    <input type="checkbox" name="sports" id="baseball">
    <label for="baseball">Baseball</label>
</form>
```

File Select box

- The file fields allow a user to browse for a local file and send it as an attachment with the form data.
- Web browsers such as Google Chrome and Firefox render a file select input field with a Browse button that enables the user to navigate the local hard drive and select a file.

```
<form>
    <label for="file-select">Upload:</label>
    <input type="file" name="upload" id="file-select">
</form>
```

Textarea

- Textarea is a multiple-line text input control that allows a user to enter more than one line of text.
- Multi-line text input controls are created using an <textarea> element.

```
<form>
   <label for="address">Address:</label>
   <textarea rows="3" cols="30" name="address" id="address"></textarea>
</form>
```

Select Boxes

- A select box is a dropdown list of options that allows user to select one or more option from a pull-down list of options.
- Select box is created using the <select> element and <option> element.
- The <option> elements within the <select> element define each list item.

Select Boxes

- **Example:**

```
<form>
    <label for="city">City:</label>
    <select name="city" id="city">
        <option value="sydney">Sydney</option>
        <option value="melbourne">Melbourne</option>
        <option value="cromwell">Cromwell</option>
    </select>
</form>
```

Submit and Reset Buttons

- A submit button is used to send the form data to a web server. When submit button is clicked the form data is sent to the file specified in the form's action attribute to process the submitted data.
- A reset button resets all the forms control to default values. Try out the following example by typing your name in the text field, and click on submit button to see it in action.

Submit and Reset Buttons

- Example:

```
<form action="action.php" method="post">
    <label for="first-name">First Name:</label>
    <input type="text" name="first-name" id="first-name">
    <input type="submit" value="Submit">
    <input type="reset" value="Reset">
</form>
```

HTML Audio/ Videos

- Sounds can be played in HTML by many different methods.
  - Using The <embed> Element

    The <embed> tag defines a container for external (non-HTML) content.

    <*embed height*="50" *width*="100" *src*="abc.mp3">
  - Using the <object> Element

    The <object> tag can also define a container for external (non-HTML) content

    <*object height*="50" *width*="100" *data*="abc.mp3"></object>

Thank You!